



Acronyms used in Astrodynamics

Here is an alphabetical non-exhaustive list of the acronyms used in the course.

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1stA	First Point of Aries Υ	CW	Clockwise
2BK	2-Body Keplerian (orbit)	DAEZ	Downrange Abort Exclusion Zone
2BN	2-Body Newtonian (orbit)	DDO	Dusk-Dawn Orbit
3LE	Three Line Elements	DE	Development Ephemerides
A&D	Aerospace and Defense	Dec	Declination
A.D.	Anno Domini / Common Era (CE)	DMS	Degrees, Minutes, Seconds
ABB	Asea Brown Boveri (company name)	DN	Descending Node Ω
ACE	Atmospheric Chemistry Experiment (SCISAT satellite payload)	DoD	Department of Defence
ADC	Analog-to-Digital Converter	DRDC	Defence Research and Development Canada
ADCS	Attitude Determination & Control System	DSN	Deep Space Network
ADU	Analog-to-Digital Units	DSO	Deep Space Object
AER	Azimuth, Elevation, Range	DST	Daylight Saving Time
AFB	Air Force Base	ECEF	Earth Centered, Earth Fixed
AFGL	Air Force Geophysics Laboratory	ECI	Earth-Centered Inertial
AFRL	Air Force Research Laboratory	ECL	Ecliptic (referential)
AIAA	American Institute of Aeronautics and Astronautics	EGM	Earth Gravitational Model
AKM	Apogee Kick Motor	ELP	Éphémérides Lunaires Parisiennes
AMD	Angular Momentum Desaturation	ENU	East/North/Up topocentric frame
AMOS	Advanced Maui Optical and Space Surveillance	EOP	Earth Orientation Parameters
AMT	Alternate Mission Target/Trajectory	EoT	Equation of Time
AN	Ascending Node Ω	ERA	Earth Rotation Angle
AOA	Abort Once Around	ESA	European Space Agency
API	Application Programming Interface	ETNO	Extreme TNO (Trans Neptunian Object)
APL	Applied Physics Laboratory	ETR	Extra-Terrestrial Radiation
ATBD	Algorithm Technical Baseline Document	ETS	Emission Transitions Spectroscopy
ATV	Automated Transfer Vehicle	FAA	(US) Federal Aviation Administration
AU	Astronomical Unit	FAI	Fédération Aéronautique Internationale
B.C.	Before Christ / Before Common Era (BCE)	FCC	Federal Communications Commission
BCE	Before Common Era	FFT	Fast Fourier Transform
CBR	Cosmic Background Radiation	FHWM	Full Width, half Maximum
CCD	Charge-Coupled Device	FIR	Far InfraRed
CCSDS	Consultative Committee for Space Data Systems	FK5	Fifth Fundamental Catalogue (historical reference)
CCW	Counterclockwise	FLI	Fraction of Lunar Illumination
CDM	Conjunction Data Message	FOR	Field of Regard
CDR	Climate Data Record / Critical Design Review	FOV	Field of View
CEO	Celestial Ephemeris Origin	FPA	Flight Path Angle (Focal Plane Array)
CEP	Celestial Ephemeris Pole	FPA	Focal Plane Array (detector)
CGS	Centimeter-gram-second system	FPGA	Field-Programmable Gate Array
CMOS	Complementary Metal-Oxide-Semiconductor	FPS	Frame Per Second
COB	Cosmic Optical Background	FSR	Free Spectral Range
CoE	Center of the Earth	FTS	Fourier Transform Spectrometer
COE	Classical Orbital Elements	FWHM	Full-Width, Half-Maximum
CR3BP	Circular-Restricted Three-Body Problem	GAIA	Global Astrometric Interferometer for Astrophysics
CRF	Celestial Reference Frame	GAST	Greenwich Apparent Sidereal Time
CS	Celestial Sphere	GCR	Galactic Cosmic Ray(s)
CSA	Canadian Space Agency	GCRS/F	Geocentric Celestial Reference System/Frame
		GCS	Geographic Coordinate System



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GECS	Geocentric-Equatorial Coordinate System	JWST	James Webb Space Telescope
GEO	Geostationary orbit	KBO	Kuiper Belt Object
GCFR	Geocentric Celestial Reference Frame	KSC	Kennedy Space Center
GCI	Geocentric Inertial	LA	Launch Azimuth
GHG	Greenhouse Gas	LAST	Local Apparent Sidereal Time
GIS	Geographic Information System	LEO	Low-Earth Orbit
GMAT	General Mission Analysis Tool (NASA)	LHA	Local Hour Angle
GMST	Greenwich Mean Sidereal Time	LLA	Latitude, Longitude, Altitude
GNSS	Global Navigation Satellite Systems	LMST	Local Mean Sidereal Time
GOES	Geostationary Operational Environmental Satellites	LN2	Liquid Nitrogen
GP	Global Perturbations	LOD	Length of Day
GPS	Global Positioning System	LOS	Line of Sight
GRC	Galactic Cosmic Rays	LOX	Liquid Oxygen
GSD	Ground Sampling Distance	LST	Local Sidereal Time
GSE	Ground Support Equipment	LTA	Linear Trajectory Assumption
GSO	Geo Synchronous Orbit	LTAN	Local Time of Ascending Node
GTO	Geostationary Transfer Orbit	LTDN	Local Time of Descending Node
HARP	High Altitude Research Project	LVLH	Local-Horizontal Local-Vertical frame
HBR	Hard Body Radius	LWIR	Long-wave Infrared
HCRS	Heliocentric ICRS-aligned system	LWST	Launch-Window Sidereal Time
HD	Henry Draper Catalog number (star catalog)	MAS	Milli Arcsecond
HEO	High-Earth Orbit / Highly-Elliptical Orbit	MECO	Main Engine Cut-Off
HIP	Hipparcos Catalog Number (star catalog)	MEO	Medium Earth Orbit
HFPA	Horizontal Flight Path Angle	MICE	MATLAB Implementation of the CSPICE Environment
HMS	Hours	MJD	Modified Julian Date
HR	Harvard Revised Number (star catalog)	MKS	Meter-kilogram-second system of units
HST	Hubble Space Telescope	MKSA	Meter-kilogram-second-ampere system of units
HWHM	Half-Width, Half-Maximum	MLI	Multi-Layer Insulation
HYG	Hipparcos, Yale, and Gliese (star catalog)	MOA	Minute of Angle
IAU	International Astronomical Union	MOC	Mission Operations Center
ICBM	Inter-Continental Ballistic Missile	MOD	Mean equator and Mean equinox of Date
ICRS/F	International Celestial Reference System/Frame	MRBM	Medium-Range Ballistic Missile
IERS	Intl. Earth Rotation & Reference Systems Service	MSL	Mean Sea Level
IFOV	Instantaneous Field of View	MST	Mean Solar Time
IJK	ECI reference frame axes	MWIR	Mid-wave Infrared
IEC	International Electrotechnical Commission	N/A	Not Applicable / Not Available
INS	Inertial Navigation Systems	NAIF	Navigation and Ancillary Information Facility (NASA)
IOC	Initial Operational Capability	NASA	National Aeronautics and Space Administration
IOD	Initial Orbit Determination	NATO	North Atlantic Treaty Organization
IRBM	Intermediate-Range Ballistic Missile	NAVSTAR	Navigation Satellite Timing and Ranging
ISA	International Standard Atmosphere	NCEI	National Centers for Environmental Information
ISO	International Organization for Standardization	NCP	North Celestial Pole
ISRO	Indian Space Research Organization	NEA	Near-Earth Asteroid
ISS	International Space Station	NED	North/East/Down topocentric frame
ITRS/F	International Terrestrial Reference System/Frame	NEP	North Ecliptic Pole
J2000	Std. reference epoch & coordinate system (2000-01-01)	NGC	New General Catalogue of Nebulae & Clusters of Stars
JAXA	Japanese Space Agency	NGSO	Non-Geostationary Orbit
JD	Julian Date	NIR	Near Infrared
JED	Julian Ephemeris Date	NIST	National Institute of Standards and Technology
JPL	Jet Propulsion Laboratory	NOAA	National Oceanic and Atmospheric Administration
JPSS	Joint Polar Satellite System	NORAD	North American Aerospace Defense Command
JSON	JavaScript Object Notation (file format)	NSSI	National Security Space Institute



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ODE	Ordinary Differential Equation	STS	Space Transportation System (Space Shuttle)
ODM	Orbit Data Message	SWAP	Size, Weight, and Power
ODPO	Orbital Debris Program Office	SWIR	Short-wave Infrared
OLR	Outgoing Longwave Radiation	SZA	Solar Zenith Angle
OMM	Orbit Mean-elements Message	TBC	To Be Completed
PEF	Pseudo-Earth-Fixed frame	TBD	To Be Defined
PM	Proper Motion	TBR	To Be Reviewed
PNE	Precession, Nutation, Equinox correction	TCA	Time to Closest Approach
POD	Precise orbit determination	TDB	Temps Dynamique Barycentrique / Barycentric Dynamical Time
PQW	Perifocal frame of reference	TEME	True Equator, Mean Equinox
PSF	Point Spread Function	TETE	True Equator, True Equinox of date
PTO	Polar Transfer Orbit	TFM	True Full Moon
PWV	Precipitable Water Vapor	TIO	Terrestrial Intermediate Origin
PYR	Pitch/Yaw/Roll	TIR	Thermal Infrared
RA	Right Ascension	TIRS	Terrestrial Intermediate Reference System / <i>That Is Rocket Science</i>
RAAN	Right Ascension of Ascending Node	TLE	Three-Line Elements (or 3LE)
RHR	Right-Hand Rule	TLWA	<i>Three-Letter Word Acronym (!...)</i>
RMS	Root Mean Square	TNO	Trans-Neptunian Object
ROIC	Read-Out Integrated Circuit	TOA	Top of Atmosphere
RSO	Resident Space Object	TOD	True of Date (coordinate system)
SAA	South Atlantic Anomaly	TOF	Time of Flight
SAR	Synthetic Aperture Radar	TPW	True Polar Wander
SC	Spacecraft	TraCCS	Traffic Coordination System for Space
SD	Serial Date	TRF	Terrestrial Reference Frame
SDA	Space Domain Awareness	TSI	Total Solar Irradiance
SDIC	Sensitivity Dependence on Initial Conditions	TST	True Solar Time
SDP	Simplified Deep Space Perturbations (e.g. SDP4)	USAF	United States Air Force (USFA Academy)
SEA	Sun Elevation Angle	USGS	United States Geological Survey
SECO	Secondary Engine Cut-off	USNO	U.S. Naval Observatory
SEE	Single Event Effect	USSF	United States Space Force
SEU	Single Event Upset	UT1	Universal Time
SEZ	South/East/Zenith topocentric frame	UTC	Coordinated Universal Time
SFU	Solar Flux Units	UTM	Universal Transverse Mercator
SGP	Simplified General Perturbations (i.e. SGP4)	VAFB	Vandenberg Air Force Base
SI	Système International	VFPA	Vertical Flight Path Angle
SLR	Satellite Laser Ranging	VIS	Visible
SLS	Space Launch System	VLBA	Very Long Baseline Array
SMAD	Space Mission Analysis and Design	VLBI	Very Long Baseline Interferometry
SNR	Signal-to-Noise Ratio	VLEO	Very Low Earth Orbit
SOFA	Standards of Fundamental Astronomy	VSOP	Variations Séculaires des Orbites Planétaires
SOI	Sphere of Influence	WGS	World Geodetic System (as WGS84)
SOL	Sun / SpaceOrbitLAB	xGEO	Beyond geosynchronous
SPICE	Spacecraft, Planet, Instrument, C-matrix (camera matrix), and Events	XYZ	ECEF reference frame axes
SRB	Solid Rocket Booster		
SRBM	Small-Range Ballistic Missile		
SRP	Solar Radiation Pressure		
SSA	Space Situational Awareness		
SSB	Solar System Barycenter		
SSO	Sun-Synchronous Orbit		
STK	Systems Tool Kit		
STM	State Transition Matrix		



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Time zones:

ADT	Atlantic Daylight Time
AEST	Australian Eastern Standard Time
AKDT	Alaska Daylight Time
AKST	Alaska Standard Time
AST	Atlantic Standard Time
BST	British Summer Time
CDT	Central Daylight Time
CET	Central European Time
CST	Central Standard Time
EDT	Eastern Daylight Time
EST	Eastern Standard Time
GMT	Greenwich Mean Time
GST	Greenwich Standard Time
HST	Hawaii Standard Time
IST	India (Navdeep)
MDT	Mountain Daylight Time
MST	Mountain Standard Time
NDT	Newfoundland Daylight Time
NST	Newfoundland Standard Time
PDT	Pacific Daylight Time
PST	Pacific Standard Time
UAE	Gulf Standard Time
WET	Western European Time

Please report any relevant suggestions for additions to this list, as well as any misspellings or erroneous designations, to:

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